

Autodesk® Scaleform®

Font and Text Configuration Overview

Scaleform 4.4
()

Scaleform C++ API

: Maxim Shemanarev, Michael Antonov
: 2.2
: 2012 6 21

Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AutoCAD LT, AutoCAD, Autodesk, the Autodesk logo, Autodesk 123D, Autodesk CAM 360, Autodesk Homestyler, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, BIM 360, Burn, Buzzsaw, CADmep, CAiCE, CAMduct, CFdesign, Civil 3D, Cleaner, Combustion, Communication Specification, Configurator 360™, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, DesignKids, DesignStudio, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWGX, DXF, Ecotect, ESTmep, Evolver, FABmep, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, ForceEffect, FormIt, Freewheel, Fusion 360, Glue, Green Building Studio, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, Incinerator, Inferno, InfraWorks, InfraWorks 360, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor HSM, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, MatchMover, Maya, Maya LT, Mechanical Desktop, MIMI, Mockup 360, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moldflow, Moondust, MotionBuilder, Movimento, MPA (design/logo), MPA, MPI (design/logo), MPX (design/logo), MPX, Mudbox, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, Productstream, Publisher 360, RasterDWG, RealDWG, ReCap, ReCap 360, Remote, Revit LT, Revit, RiverCAD, Robot, Scaleform, Showcase, Showcase 360 ShowMotion, Sim 360, SketchBook, Smoke, Socialcam, Softimage, Sparks, SteeringWheels, Stitcher, Stone, StormNET, TinkerBox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, ViewCube, Visual LISP, Visual, VRED, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

Overview)
Autodesk Scaleform Corporation
6305 Ivy Lane, Suite 310
Greenbelt, MD 20770, USA
www.scaleform.com
info@scaleform.com
(301) 446-3200
(301) 446-3199

FAX

		1
1.1	Flash	2
1.1.1		2
1.1.2		3
1.1.3		4
1.1.4		5
1.2	UI	6
		8
2.1		8
2.1.1		9
2.1.2		11
2.2	gfxfontlib.swf	11
		13
3.1		13
3.1.1		15
3.1.2	Scaleform Player	17
3.1.3		18
3.1.4		19
3.2		21
		23
4.1		23
4.2	Gfx::FontMap	25
4.3	Gfx::FontLib	26
4.4	Gfx::FontProviderWin32	26
4.4.1		27
4.4.2		28
4.5	Gfx::FontProviderFT2	29
4.5.1	FreeType	30
4.5.2		30
		33
5.1		34

5.2		35
5.3	gfxexport	38
5.4	- gfxexport	38
5.5		40
5.6		42
		44
6.1		44
6.2		46
6.3		47
6.4	ActionScript	47

HTML

GFX/SWF

FreeType 2

/

/

Scaleform

Flash® Studio
SWF

Flash Studio

Flash

Flash

Scaleform

GfX::Translator

GfX::FontLib

FreeType 2

Scaleform

Flash Studio Scaleform
Flash

Scaleform

Flash

Scaleform

UI

Flash
Flash

Flash

(Text Field)

UI

Flash

UI

のテキストとフォントの基礎

Flash Studio

UI

Flash Studio

Flash

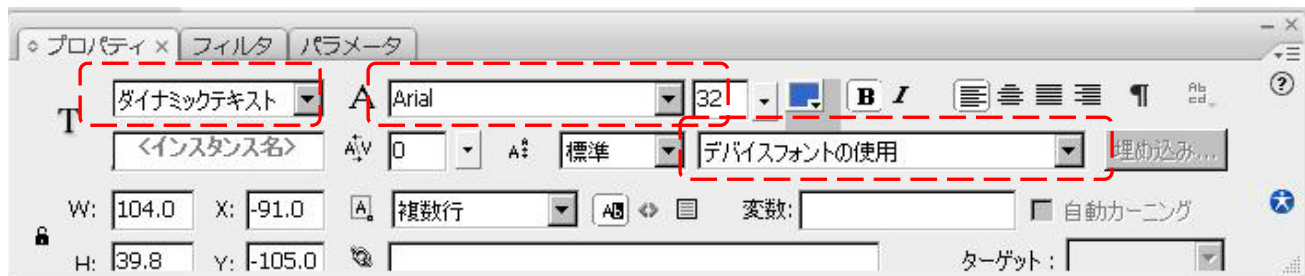
GF

1 -

Flash

Flash Studio

[]



ActionScript

GfX::Translator

2

(

)

a)

b)

Arial

[

]

[

]

SWF

Flash 8

(

)

(

)

(

)

1

SWF

Flash Player

Scaleform

(

)

(

)

[

...]

1

TextField

textfield

(Scaleform

[

GFx::FontLib

...]

[

)

]

1			1 KB	450 K
114	-	+	12 KB	480 K
596	-		70 KB	630 K
7,583	-		2,089 KB	3,500 K
18,437	-		5,131 KB	8,000 K

“Arial Unicode MS”

500

150K

1. UI

Flash

(

2. Gfx::FontLib

(

)
SWF/GFX

Gfx::FontLib

3.

IME

4. GfxEport

-fc
10 30%

5.3

Gfx::FontLib

IME

ゲームのフォントに関する決定

Scaleform

UI

4

1:

2:

3:

Scaleform

4:

Scaleform

UI

UI

2

3

読み込みフォントの置き換え

Gfx::FontLib

Player

デバイスフォントエミュレーション

()

カスタムアセットの生成 Flash

SWF/GFX

Scaleform

3:

Scaleform

Gfx::FontLib

FreeType 2

Gfx::FontLib

4

Scaleform

3

gfxexport

UI

UI

UI

UI

Scaleform

fonts_en.swf fonts_kr.swf

SWF

Flash

Flash

フォントシンボル

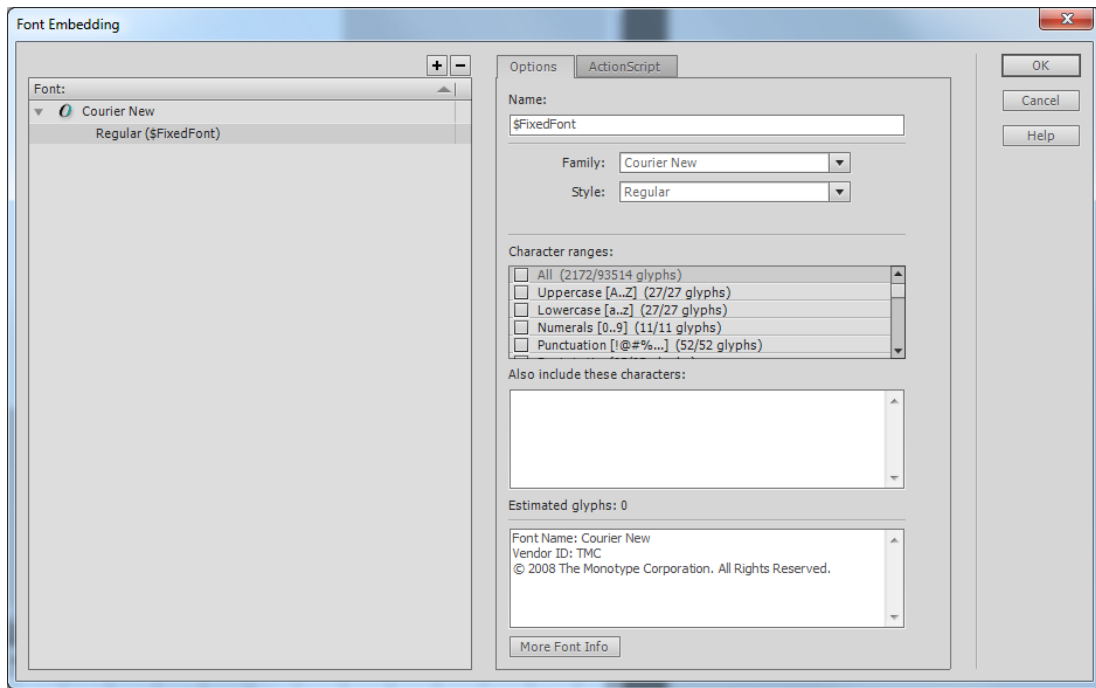
W

Flash Studio

L

[

...]



FLA

“\$FixedFont”

“\$ArialGame*”

UI

Web

]

-
Flash Studio [

\$

FLA

/

FLA

[]

[]

FLA

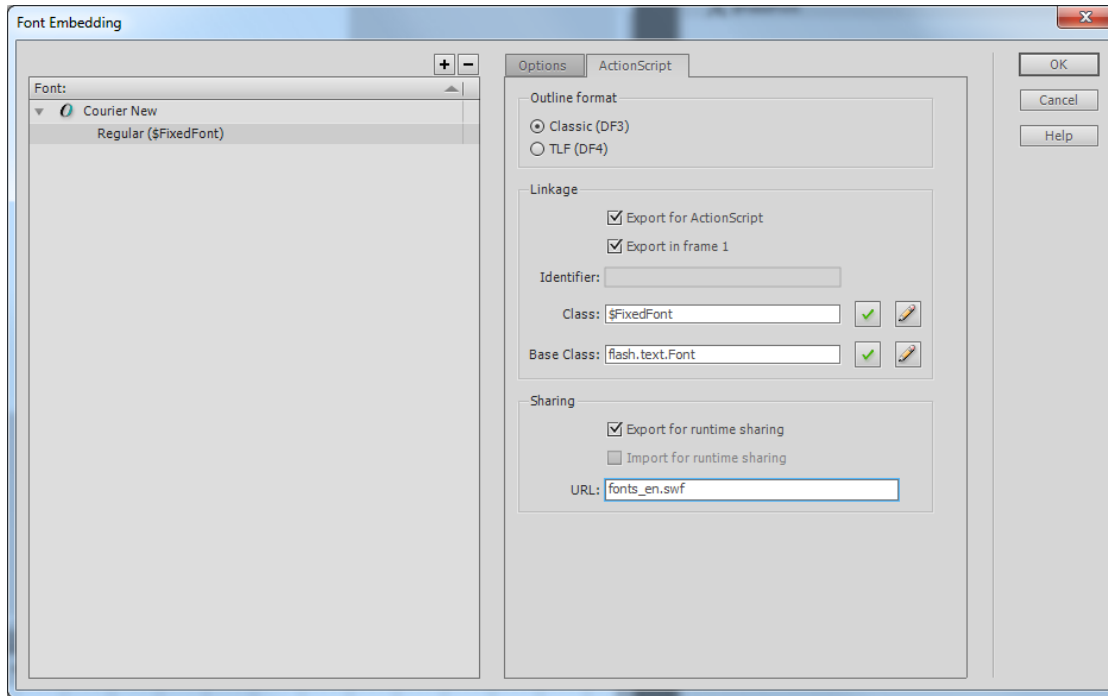
FLA

Flash

/

"Properties"

"ActionScript"



"OK"

Flash

ActionScript

URL

SWF

GfX::FontLib

URL

"fonts_en.swf"

```
Loader.SetDefaultFontLibName("fonts_en.swf");
```

```
GfXPlayer  
fontlib "fonts_en.swf"
```

fonconfig.txt

fontlib

" : \$ "

/

Scaleform

SWF

SWF

```

        TextField.htmlText
        ("Arial"
        )
        ActionScript
    (
    )
    TextFormat

```

Flash Studio CS4

```

    Windows
    [Unicode
        243
    ]
    [
        "English"
    ] > [
        11,920
    ] >
    "gfxfontlib.swf"

```

```

    Flash CS5
    "gfxfontlib.swf"
    fonts_en.swf fonts_jp.swf

```

```

    Flash Studio CS4
    "gfxfontlib.swf"

```

GFx 4.0

を作成する手順

Flash Studio FLA

1.

2. FLA

3. 1

4.

[]

5.

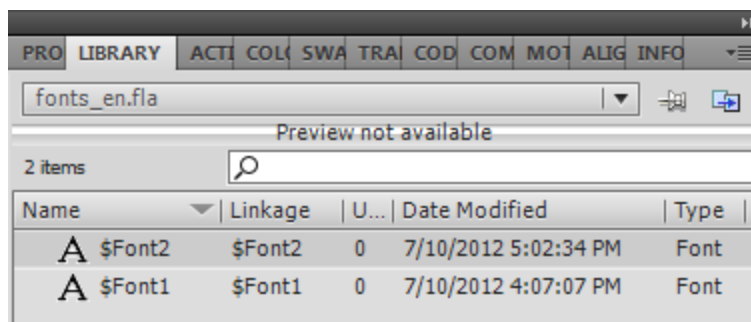
6.

fonts_<lang>.swf

lang

FontMap

FLA



/

UI

7. 読み込みフォントの置き換え
8. デバイスフォントエミュレーション
9. カスタムアセットの生成

Gfx::FontLib

Gfx::FontMap

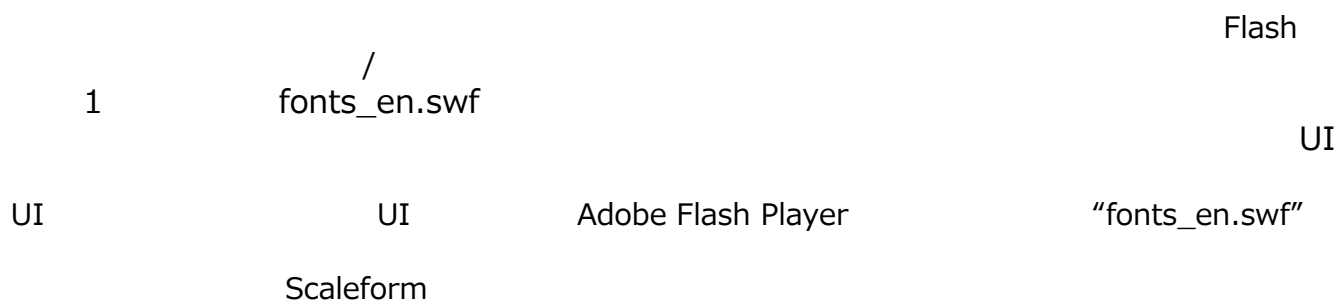
Gfx::Translator

Gfx::Translator
UI

Flash Studio

UI

読み込みフォントの置き換え



Loader::SetDefaultFontLibName (const char* filename)

C++

```
Loader.SetDefaultFontLibName("fonts_kr.swf");
```

SWF

CreateInstance

1. 文字列変換テーブル: ゲームの各フレーズを翻訳されたフレーズにマップするものです。

2. :

3. : ("NormalFont")

Scaleform

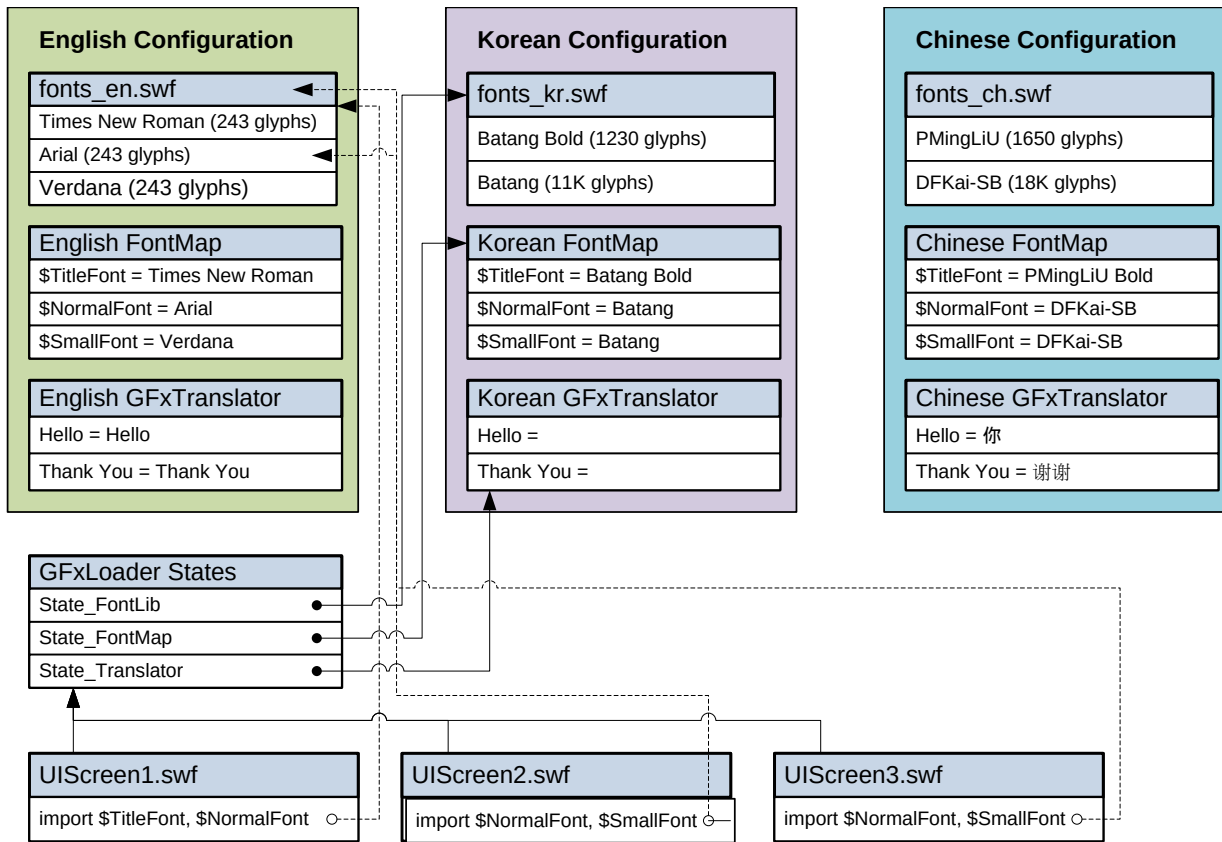
Gfx::Loader

Gfx::Translator Gfx::FontLib

Gfx::FontMap

Gfx::Loader::CreateMovie

CreateMovie



"UIScreen1.swf" "UIScreen3.swf" "fonts_en.swf"

Scaleform
 GFLoader
 GFLoader::SetDefaultFontLibName()
 GFLoader::FontLib
 GFLoader::FontMap
 GFLoader::Loader
 2
 1
 2
 UI

"fonts_kr.swf"

UI

		SWF	
	GFX		
GfX::FontMap		GfX::Translator	
Scaleform		"fontconfig.txt"	
1.	1		"fonts_en.swf"
2.			
	GfX::Translator		
3.			GfX::FontMap
4.		"fonts_kr.swf"	
		UI	
1.		GfX::Loader	
	GfX::Translator	Translate	
2.	GfX::FontMap	GfX::Loader	MapFont
		GfX::FontMap	
	GfX::FontMap	3	
3.		(	
	GfX::FontCacheMamanger, GfX::FontProvider	GfX::FontPackParams	)
4.	GfX::FontLib	GfX::Loader	
5.		SetDefaultFontLibName(filename)	
	swf		
	Loader.SetDefaultFontLibName("fonts_en.swf");		
	GfX::FontLib		SWF/GFX
		GfX::FontLib::AddFromFile	
6.	GfX::Loader::CreateMovie		

fontconfig.txt
Scaleform Player
)
SWF/GFX
(
Scaleform Player
)

F2
Ctrl+N
Scaleform Player HUD

8
ASCII
UTF-16

```
[FontConfig "English"]
fontlib "fonts_en.swf"
map "$TitleFont" = "Times New Roman" Normal
map "$NormalFont" = "Arial " Normal
map "$SmallFont" = "Verdana" Normal
```

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$TitleFont" = "Batang" Bold
map "$NormalFont" = "Batang" Normal
map "$SmallFont" = "Batang" Normal
tr "Hello" = "안녕하세요"
tr "Thank You" = "감사합니다"
```

```
[FontConfig "Chinese"]
fontlib "fonts_ch.swf"
map "$TitleFont" = "PMingLiU" Bold
map "$NormalFont" = "DFKai-SB" Normal
map "$SmallFont" = "DFKai-SB" Normal
tr "Hello" = "你 "
tr "Thank You" = "谢谢"
```

name
Scaleform Player HUD
[FontConfig

fontlib "fontfile"	SWF/GFX fontlib Gfx::FontLib
map "\$UIFont" = "PMingLiU"	Gfx::FontMap UI Gfx::FontMap \$UIFont "fonts_en.swf" UI

TextFormat

Flash

```
("$TitleFont"    )
```

("Arial")

"fonts_en.swf"

“fonts_en.swf”

"fonts_en.swf"

UI

2

fontconfig.txt

SWF

1. FLA ActionScript 2.0

3.0

SWF

2. FLA

ActionScript

3.

\$Font1

“Family”

"Arial"

“Style”

“Regular”

4. "Character ranges"

'Basic Latin'

5.

for run-

OK

ActionScript 3.0 Flash Studio

6.

3 5

/

Flash Studio 2

7. 1 FLA

ActionScript

8.

\$Font1

"Family"

" " "Style"

"Regular"

9. "Character ranges"

10.

Export

for run-

OK

ActionScript 3.0 Flash Studio

11.

8 10

12.

13. main-app.fla.

Ctrl-C

14.

-

j

Ctrl-V

15.

-

ID

16. 2

17.

-

18.

swf

fontconfig.txt

Unicode

UTF-8

```
[FontConfig "English"]
fontlib "fonts_en.swf"
map "$Font1" = "Arial"
map "$Font2" = "Times New Roman"
tr "$TEXT1" = "This is"
tr "$TEXT2" = "ENGLISH!"
```

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$Font1" = "Batang"
map "$Font2" = "BatangChe"
tr "$TEXT1" = " "
tr "$TEXT2" = " !"
```

重要 :

“ “ “ “
 “Batang” “BatangChe”
 SWF

Flash SWF
 fontconfig.txt

gfxexport -fntlst <swfname>

*.lst

19.

GFxPlayer

-app.
 Ctrl-N

Unicode

UTF-8

fonts_kr.swf

fonts_kr.swf

カスタムアセットの生成

SWF/GFX

“UIScreen.fla”
 SWF

“UIScreen.swf”

“UIScreen.swf”

(UI 600K

)

UI

UI

()

UI

IME

Flash Studio

[

]

(Ctrl + F11)

Scaleform

Flash

[

]

Flash
HTML

1. ローカルに埋め込まれたフォント

2. SWF/GFX

3. GfX::FontLib SWF/GFX

4. (GfX::FontProviderWin32)

Flash

GfX::FontMap GfX::FontProvider

GfX::FontLib

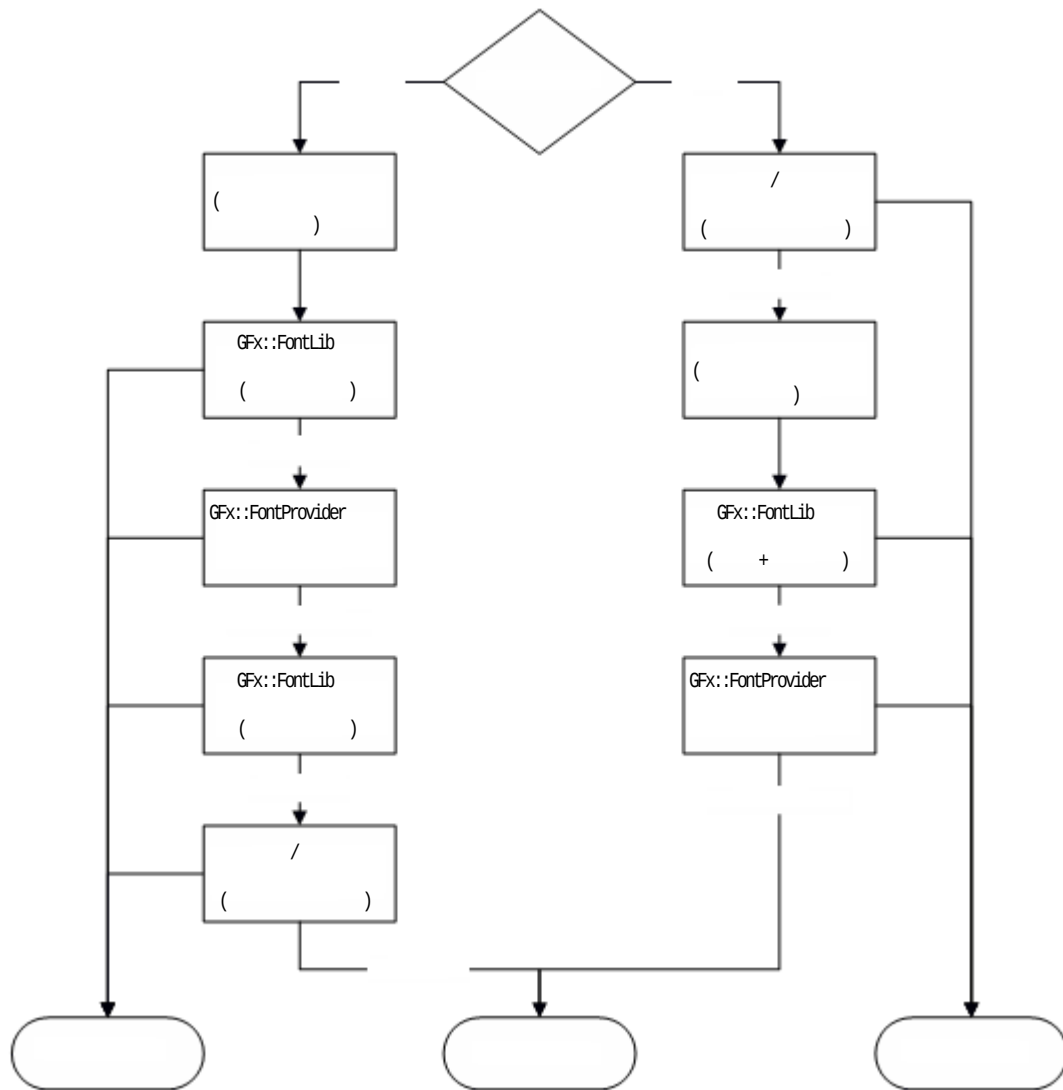
GfX::FontLib
GfX::FontMap
SWF/GFX

SWF GfX::Loader::SetFontProvider

SWF
Scaleform

GfX::FontProviderWin32 – Win32 API

GfX::FontProviderFT



Gfx::FontLib Gfx::FontProvider

GfX::FontMap

3

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$TitleFont" = "Batang" Bold
map "$NormalFont" = "Batang" Normal
map "$SmallFont" = "Batang" Normal
```

("\$TitleFont")
)

(

C++

```
#include "GfX/GfX_FontLib.h"
. . .
Ptr<FontMap> pfontMap = *new FontMap;
Loader.SetFontMap(pfontMap);

pfontMap->MapFont("$TitleFont", "Batang", FontMap::MFF_Bold, scaleFactor = 1.0f);
pfontMap->MapFont("$NormalFont", "Batang", FontMap::MFF_Normal, scaleFactor = 1.0f);
pfontMap->MapFont("$SmallFont", "Batang", FontMap::MFF_Normal, scaleFactor = 1.0f);
```

"\$NormalFont"

"\$SmallFont"

3

MapFont

MFF_Original

scaleFactor
scaleFactor 1.0f

GfX::FontMap

GfX::Loader::CreateMovie

GfX::MovieDef

3	GFx::FontLib	2	1	"fonts_en.swf"
---	--------------	---	---	----------------

```
#include "GFx/GFx_FontLib.h"
. . .
Ptr<FontLib> fontLib = *new FontLib;
Loader.SetFontLib(fontLib);
fontLib->SetSubstitute("<default_font_lib_swf_or_gfx_file>");

Ptr<MovieDef> m1 = *Loader.CreateMovie("<swf_or_gfx_file1>");
Ptr<MovieDef> m2 = *Loader.CreateMovie("<swf_or_gfx_file2>");
. . .
fontLib->AddFontsFrom(m1, true);
fontLib->AddFontsFrom(m2, true);
```

AddFontsFrom			AddFontsFrom
2	(	m1	AddRef m2) (false)
			/

GFx::FontLib

GFx::FontProviderWin32	Win32 API	GetGlyphOutline()
------------------------	-----------	-------------------

```
#include "GFx/GFx_FontProviderWin32.h"
. . .
Ptr<FontProviderWin32> fontProvider = *new FontProviderWin32(::GetDC(0));
Loader.SetFontProvider(fontProvider);
```

GFx::FontProviderWin32 (DC)	DC	Windows	(::GetDC(0))
--------------------------------	----	---------	--------------

Scaleform
CJK (Chinese /

Japanese / Korean)
CJK
CJK

API

Gfx::FontProviderWin32 Gfx::FontProviderFT2

4

```
Font::NativeHintingRange vectorRange;  
Font::NativeHintingRange rasterRange;  
unsigned maxVectorHintedSize;  
unsigned maxRasterHintedSize;
```

vectorRange rasterRange

```
Font::DontHint -  
Font::HintCJK -  
Font::HintAll -
```

rasterRange vectorRange maxVectorHintedSize maxRasterHintedSize

SimSun

abcdefghijklmnopqrstuvwxyz vectorRange=GfxFont::DontHint
ABCDEFGHIJKLMNOPQRSTUVWXYZ rasterRange=GfxFont::DontHint

僞偽僞僞
僞僞

abcdefghijklmnopqrstuvwxyz vectorRange=GfxFont::HintAll
ABCDEFGHIJKLMNOPQRSTUVWXYZ rasterRange=GfxFont::DontHint

僞偽僞僞僞
僞僞

abcdefghijklmnopqrstuvwxyz vectorRange=GfxFont::HintAll
ABCDEFGHIJKLMNOPQRSTUVWXYZ rasterRange=GfxFont::HintCJK

僞偽僞僞僞
僞僞

“Arial Unicode MS ”

abcdefghijklmnopqrstuvwxyz
ABCDEFGHIJKLMNOPQRSTUVWXYZ
僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞
僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞

Adobe FLASH Scaleform

Win32 FreeType

```
vectorRange = Font::DontHint;  
rasterRange = Font::HintCJK;  
maxVectorHintedSize = 24;  
maxRasterHintedSize = 24;
```

GFx::FontProviderWin32 CJK

```
fontProvider->SetHintingAllFonts(. . .);
```

or

```
fontProvider->SetHinting(fontName, . . .);
```

SetHintingAllFonts () SetHinting() ()

```
void SetHintingAllFonts(Font::NativeHintingRange vectorRange,  
                         Font::NativeHintingRange rasterRange,  
                         unsigned maxVectorHintedSize=24,  
                         unsigned maxRasterHintedSize=24);
```

```
void SetHinting(const char* Name,
                Font::NativeHintingRange vectorRange,
                Font::NativeHintingRange rasterRange,
                unsigned maxVectorHintedSize=24,
                unsigned maxRasterHintedSize=24);
```

Name UTF-8

David Turner FreeType-2

Gfx::FontProviderWin32

FreeType

()

Windows MSVC

```
freetype<ver>.lib - DLL
freetype<ver>_D.lib - DLL
freetype<ver>MT.lib - ( CRT)
freetype<ver>MT_D.lib - ( CRT)

"<ver>" FreeType ( 2.1.9 219)

freetype2/include freetype2/objs
```

Gfx::FontProviderFT2

```
#include "Gfx/Gfx_FontProviderFT2.h"
. . .
Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
<Map Font to Files or Memory>
Loader.SetFontProvider(fontProvider);
```

FontProviderFT2(FT_Library lib=0);

FreeType ()
FreeType FreeType Scaleform

(FT_Done_FreeType)

Gfx::Loader (FreeType

fontProvider->GetFT_Library() FT_Library
FreeType
Gfx::FontProviderFT2 FreeType

Win32 API FreeType ()

Gfx::FontProviderFT2

```
void MapFontToFile(const char* fontName, unsigned fontFlags,  
                  const char* fileName, unsigned faceIndex=0  
                  Font::NativeHintingRange vectorHintingRange = Font::DontHint,  
                  Font::NativeHintingRange rasterHintingRange = Font::HintCJK,  
                  unsigned maxVectorHintedSize=24,  
                  unsigned maxRasterHintedSize=24);
```

```
"fontName" ( "Times New Roman")  
"fileName" ( "C:\\WINDOWS\\Fonts\\times.ttf")  
"fontFlags" "Font::FF_Bold" "Font::FF_Italic" "Font::FF_BoldItalic"  
"Font::FF_BoldItalic"  
"Font::FF_Bold | Font::FF_Italic" "faceIndex" FT_New_Face  
0
```

Win32 FreeType

FreeType

```
void MapFontToMemory(const char* fontName, unsigned fontFlags,  
                    const char* fontData, unsigned dataSize, unsigned faceIndex=0  
                    Font::NativeHintingRange vectorHintingRange = Font::DontHint,  
                    Font::NativeHintingRange rasterHintingRange = Font::HintCJK,  
                    unsigned maxVectorHintedSize=24,  
                    unsigned maxRasterHintedSize=24);
```

```

FILE* fd = fopen("C:\\WINDOWS\\Fonts\\times.ttf", "rb");
if (fd) {
    fseek(fd, 0, SEEK_END);
    unsigned size = ftell(fd);
    fseek(fd, 0, SEEK_SET);
    char* font = (char*)malloc(size);
    fread(font, size, 1, fd);
    fclose(fd);
    fontProvider->MapFontToMemory("Times New Roman", 0, font, size);
}

```

()

Gfx::Loader

FreeType

Scaleform

Windows FreeType

```

Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
fontProvider->MapFontToFile("Times New Roman", 0,
    "C:\\WINDOWS\\Fonts\\times.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Bold,
    "C:\\WINDOWS\\Fonts\\timesbd.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Italic,
    "C:\\WINDOWS\\Fonts\\timesi.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_BoldItalic,
    "C:\\WINDOWS\\Fonts\\timesbi.ttf");

fontProvider->MapFontToFile("Arial", 0,
    "C:\\WINDOWS\\Fonts\\arial.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Bold,
    "C:\\WINDOWS\\Fonts\\arialbd.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Italic,
    "C:\\WINDOWS\\Fonts\\ariali.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_BoldItalic,
    "C:\\WINDOWS\\Fonts\\arialbi.ttf");

fontProvider->MapFontToFile("Verdana", 0,
    "C:\\WINDOWS\\Fonts\\verdana.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Bold,
    "C:\\WINDOWS\\Fonts\\verdanab.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Italic,
    "C:\\WINDOWS\\Fonts\\verdanai.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_BoldItalic,

```

```
        "C:\\WINDOWS\\Fonts\\verdanaz.ttf");  
    . . .  
    Loader.SetFontProvider(fontProvider);
```

MapFontToFile()

Scaleform (1 2)

- 1.
- 2. (mip-map gfxexport)
- 3. (Gfx::FontPackParams)
- 4.

PC Xbox 360 PS3 Wii

Gfx::FontProviderWin32 Gfx::FontProviderFT2

gfxexport CPU
(PSP PS2)

Scaleform

フォントキャッシュマネージャ

Scaleform

Render::GlyphCacheConfig

Gfx::FontPackParams
Renderer2D

NULL

GFX

		Scaleform	3.x	4.0
Gfx 3.3	GfxLoader	GfxFontCacheManager		
		Gfx 4.0	Renderer2D	
	GlyphCacheConfig			Scaleform Gfx 4.0
		5.2		

テクスチャからラスターグリフがレンダーされる時は、グリフキャッシュシステムは必ず使用されます。

Gfx::FontPackParams

()

```
bool Done = false;

if (Font has Static Textures with Packed Glyphs)
{
    if (GlyphSize <
        FontPackParams.TextureConfig.NominalSize * MaxRasterScale)
    {
        Draw the Glyph as a Texture using Static Cache;
        Done = true;
    }
}
```

```

    }
}
else if (Dynamic Cache is Enabled AND
        GlyphSize < GlyphCacheParams.MaxSlotHeight)
{
    Draw the Glyph as a Texture using Dynamic Cache;
    Done = True;
}

if (Not Done)
{
    Draw the Glyph as Vector Shape;
}

```

ダイナミックフォントキャッシュの使用

(10000)

デフォルトではダイナミックキャッシュはイネーブルされ、そのバッファを割り当てます。

```
renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));
```

renderer

Render::Renderer2D

HAL

。

LRU (Least Recently Used)


```

Render::GlyphCacheParams gcparams;
gcparams.TextureWidth    = 1024;
gcparams.TextureHeight   = 1024;
gcparams.MaxNumTextures  = 1;
gcparams.MaxSlotHeight   = 48;
gcparams.SlotPadding     = 2;
gcparams.TexUpdWidth     = 256;
gcparams.TexUpdHeight    = 512;

renderer->GetGlyphCacheConfig()->SetParams(gcparams);

```

TextureWidth, TextureHeight -
2

MaxNumTextures -

MaxSlotHeight -

SlotPadding -
2

TexUpdWidth, TexUpdWidth -
256x512 (128K)) 256x256 128x128

LRU
1MB 4KB 4 K

/ 256 4 K 16

4K 100 256 4K 1 MB

256 2

MaxSlotHeight+2*SlotPadding 2

2-5

(60-80%

(UI 500 2000

1

) (Flash () (

GFX, Dynamic cache	Readability	Animation
<i>Anti-alias for readability</i> <i>Anti-alias for animation</i>		

(()) Scaleform Flash

Scaleform

(O G C Q o g e)

Scaleform

H E F T U V W X Z
Z X V W

1

1
Scaleform

[] Flash "Zz"

フォントのコンパクター、の使用

GFX gfxexport SWF
 -fc DDS SWF
 TGA
 gfxreport

-fc	
-fcl <size>	256 Flash 1024 25% 256
-fcm	FontCompactor 10 70% 32 64 12 16 32 16 64

フォント テクスチャの前処理

gfxexport -fonts

Scaleform Player GFX

gfxexport

CPU

Gfx::ImageCreator

Render::Renderer

mip-map ()

gfxexport -fonts -strip_font_shapes test.swf

strip_font_shapes Scaleform

gfxexport

gfxexport

-fonts	()
-fns <size>	48 () 1 mip-map
-fpp <n>	3
-fts <WxH>	256x256 ('-fts 128' 128x128 1) '-fts 512x128'
-fs	

-strip_font_shapes	GFX
-fi <format>	TGA8 (grayscaled) TGA24 (grayscaled) TGA32 (-i) A8 TGA8 DDS <format> DDS8 TGA DDS

5.2

フォント グリフ パッカーの設定

GFXExport

(
)

Scaleform

```

Ptr<FontPackParams> packParams = *new FontPackParams();
Loader.SetFontPackParams(packParams);

```

GFXExport (
.Scaleform
)

GFXExport

(
)

```

Loader.GetFontPackParams()->SetUseSeparateTextures(Bool flag);
Loader.GetFontPackParams()->SetGlyphCountLimit(int lim);
Loader.GetFontPackParams()->SetTextureConfig(fontPackConfig);

```

SetUseSeparateTextures()
"true"

"false"

SetGlyphCountLimit()

0 ()

500

)

SetTextureConfig()

```
FontPackParams::TextureConfig fontPackConfig;
fontPackConfig.NominalSize = 48;
fontPackConfig.PadPixels = 3;
fontPackConfig.TextureWidth = 1024;
fontPackConfig.TextureHeight = 1024;
```

NominalSize-
)

(

RAM

"NominalSize"

NominalSize

(

PadPixels -

1

TextureWidth TextureHeight -

2

1

.GFX

```
2 Ptr<FontPackParams> packParams = *new FontPackParams();
  Loader.SetFontPackParams(packParams);
  ...
  renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));
```

```
3 Ptr<FontPackParams> packParams = *new
  FontPackParams(); Loader.SetFontPackParams(packParams);
  Loader.GetFontCacheManager()->EnableDynamicCache(false);
```

()

ベクトル化の制御

Scaleform

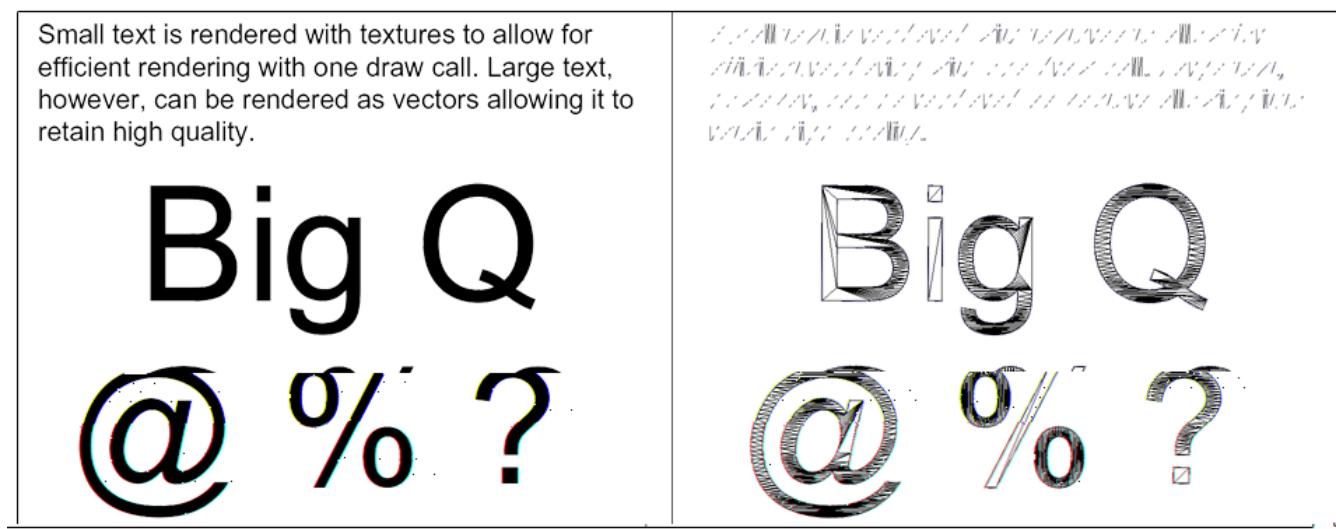
Flash

1

Scaleform

Scaleform Player

Ctrl+W



(MaxSlotHeight)
MaxSlotHeight

Render::GlyphCacheConfig::SetParams

(mip-maps)

GlyphCacheParams:: MaxRasterScale
SetMaxRasterScale

()

MaxRasterScale 1 1.25
1.25

Player
48

60

MaxRasterScale

strip_font_shapes

gfxexport

GFX

Scaleform

Player

Scaleform
Adobe Flash

Flash

Flash

Scaleform

Scaleform

フィルタの種類、使用可能なオプションと制約

Flash Studio

1



Flash

Scaleform

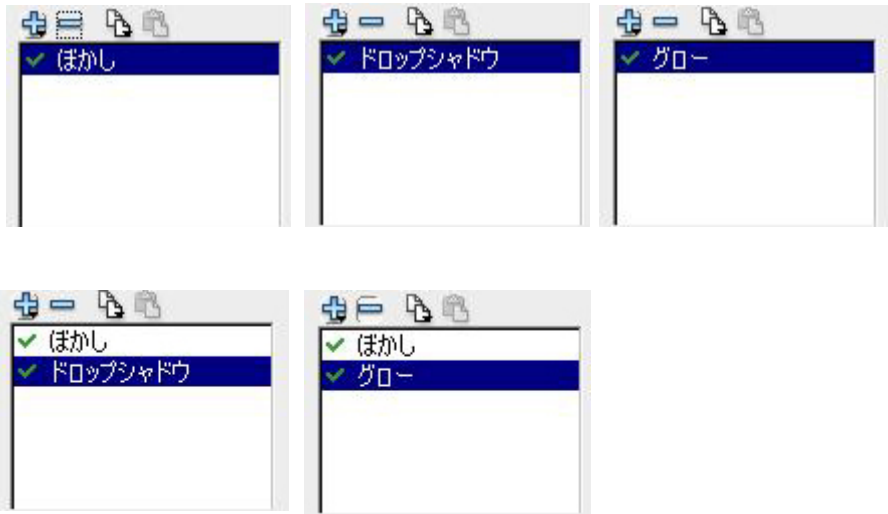
Scaleform
Flash

Scaleform

Flash
Scaleform

-
-
-
-

-



Flash

$$\left[\begin{pmatrix} \text{Blur X} \\ \text{Blur Y} \end{pmatrix} \right] \left(\begin{pmatrix} \text{Scaleform} \\ \text{Scaleform} \end{pmatrix} \right)^2 \left[\begin{pmatrix} \text{Flash} \\ \text{Scaleform} \end{pmatrix} \right]$$

Flash

Text

GFx

Text

$$\left[\begin{pmatrix} \text{Flash} \\ \text{GFx} \end{pmatrix} \right] \left(\begin{pmatrix} \text{Flash} \\ \text{Scaleform} \end{pmatrix} \right)$$

フィルタ品質

Flash

3x3
2

3x3
[]

3

[]
[] []
[] []
[]

Flash

Scaleform

Scaleform

[]
[:]

[]

()

Scaleform

(

)

	Low quality	Medium quality	High quality
Flash	<i>Text</i>	<i>Text</i>	<i>Text</i>
GFx	<i>Text</i>	<i>Text</i>	<i>Text</i>
	Flash	Scaleform	Scaleform

Smoothin.html

<http://www.ph.tn.tudelft.nl/Courses/FIP/noframes/fip->

フィルタのアニメーション

Flash

Scaleform ActionScript

Scaleform

()
()

からのフィルターの使用

Scaleform AS2 AS3 /
DropShadowFilter BlurFilter ColorMatrixFilter BevelFilter
Flash